

CS 6371 - Assignment 6

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1. (a)

$$\begin{aligned}
 e &= (\text{or } (\text{not true}) \text{ false}) \\
 &= ((\lambda a. \lambda b. (a ? \text{true} : b)) (\text{not true}) \text{ false}) \\
 &= ((\lambda a. \lambda b. ((a)(\text{true})(b))) (\text{not true}) \text{ false}) \\
 &= ((\lambda a. \lambda b. ((a)(\text{true})(b))) ((\lambda b. (b ? \text{false} : \text{true})) \text{true}) \text{false}) \\
 &= ((\lambda a. \lambda b. ((a)(\text{true})(b))) ((\lambda b. ((b)(\text{false})(\text{true}))) \text{true}) \text{false}) \\
 &= ((\lambda a. \lambda b. ((a)(\lambda x. \lambda y. x)(b))) ((\lambda b. ((b)(\text{false})(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) \text{false}) \\
 &= ((\lambda a. \lambda b. ((a)(\lambda x. \lambda y. x)(b))) ((\lambda b. ((b)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y))
 \end{aligned} \tag{1}$$

(b)

$$\begin{aligned}
 &((\lambda a. \lambda b. ((a)(\lambda x. \lambda y. x)(b))) ((\lambda b. ((b)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &((\lambda b. (((\lambda b. ((b)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. x)(b))) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &(((\lambda b. ((b)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x))) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. x) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &((((\lambda x. \lambda y. x)(\lambda x. \lambda y. y)(\lambda x. \lambda y. x)) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &((((\lambda y. (\lambda x. \lambda y. y)) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &(((\lambda x. \lambda y. y) (\lambda x. \lambda y. x)) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &((\lambda y. y) (\lambda x. \lambda y. y)) \\
 \rightarrow_1 &(\lambda x. \lambda y. y)
 \end{aligned} \tag{2}$$

2. (a)

$$isempty = \lambda \ell. \text{not } (\pi_1 \ell) \tag{3}$$

(b)

$$\begin{aligned}
 head &= \lambda \ell. \pi_1 (\pi_2 \ell) \\
 tail &= \lambda \ell. \pi_2 (\pi_2 \ell)
 \end{aligned} \tag{4}$$

(c)

$$cons = \lambda e. \lambda \ell. (\text{pair true } (\text{pair } e \ell)) \tag{5}$$

(d)

$$length = Y(\lambda f. \lambda \ell. (isempty \ell ? 0_{\mathbb{N}} : succ_{\mathbb{N}}(f (tail \ell)))) \tag{6}$$

(e)

$$\begin{aligned}
rev &= Y(\lambda f.(\lambda a.\lambda \ell.(\text{isempty } \ell ? [] : f ((\text{head } \ell) :: a) (\text{tail } \ell)))) ([]) \\
append_helper &= Y(\lambda f.(\lambda \ell_1.\lambda \ell_2.(\text{isempty } \ell_2 ? \ell_1 : f ((\text{head } \ell_2) :: \ell_1) (\text{tail } \ell_2)))) \\
append &= \lambda \ell_1.\lambda \ell_2.(append_helper \ell_2 (rev \ell_1))
\end{aligned} \tag{7}$$

(f)

$$map = Y(\lambda m.\lambda a.\lambda f.\lambda \ell.(\text{isempty } \ell ? a : m ((f (\text{head } \ell)) :: a) f (\text{tail } \ell))) ([]) \tag{8}$$

(g)

$$fold_left = Y(\lambda g.\lambda f.\lambda a.\lambda \ell.(\text{isempty } \ell ? a : g (f) (f a (\text{head } \ell)) (\text{tail } \ell))) \tag{9}$$

3. **Theorem.** If $e \Downarrow e''$ and $e \rightarrow_1 e'$ then $e' \Downarrow e''$.

Proof. There exists a derivation D_L of $e \Downarrow e''$ and a derivation D_S of $e \rightarrow_1 e'$.

We will prove $e' \Downarrow e''$ by structural induction on D_S .

Inductive Hypothesis. If $e_0 \Downarrow e_0''$ has a derivation smaller than D_L and $e_0 \rightarrow_1 e_0'$ has a derivation smaller than D_S , then $e_0' \Downarrow e_0''$.

Inductive Cases.

Case 1. Suppose $D_S = (\lambda v.e_1)e_2 \rightarrow_1 e_1[e_2/v]$ and $D_L = (\lambda v.e_1)e_2 \Downarrow e''$.

We must prove $e_1[e_2/v] \Downarrow e''$.

D_L has derivation:

$$\frac{\lambda v.e_1 \Downarrow \lambda v.e_1 \quad e_1[e_2/v] \Downarrow e''}{(\lambda v.e_1)e_2 \Downarrow e''} \tag{10}$$

By assumption from D_L , $e_1[e_2/v] \Downarrow e''$.

Case 2. Suppose:

$$\begin{aligned}
D_S &= \frac{\frac{D_0}{e_1 \rightarrow_1 e_1'}}{e_1 e_2 \rightarrow_1 e_1' e_2} \\
D_L &= \frac{\frac{D_1}{e_1 \Downarrow \lambda v.e} \quad \frac{D_2}{e[e_2/v] \Downarrow e''}}{e_1 e_2 \Downarrow e''}
\end{aligned} \tag{11}$$

In order to prove $e_1' e_2 \Downarrow e''$, it is sufficient to prove it's derivation.

$$\frac{\frac{D'_1}{e_1' \Downarrow \lambda v.e} \quad \frac{D'_2}{e[e_2/v] \Downarrow e''}}{e_1' e_2 \Downarrow e''} \tag{12}$$

We apply the inductive hypothesis with $D_0 < D_S$, $D_1 < D_L$, $e_0 = e_1$, $e'_0 = e'_1$, $e''_0 = \lambda v.e.$. Therefore, $e'_1 \Downarrow \lambda v.e.$ D'_1 and D'_2 are proven, thus $e'_1 e_2 \Downarrow e''$.

D'_2 by assumption D_2 . $\square$